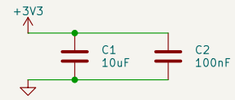
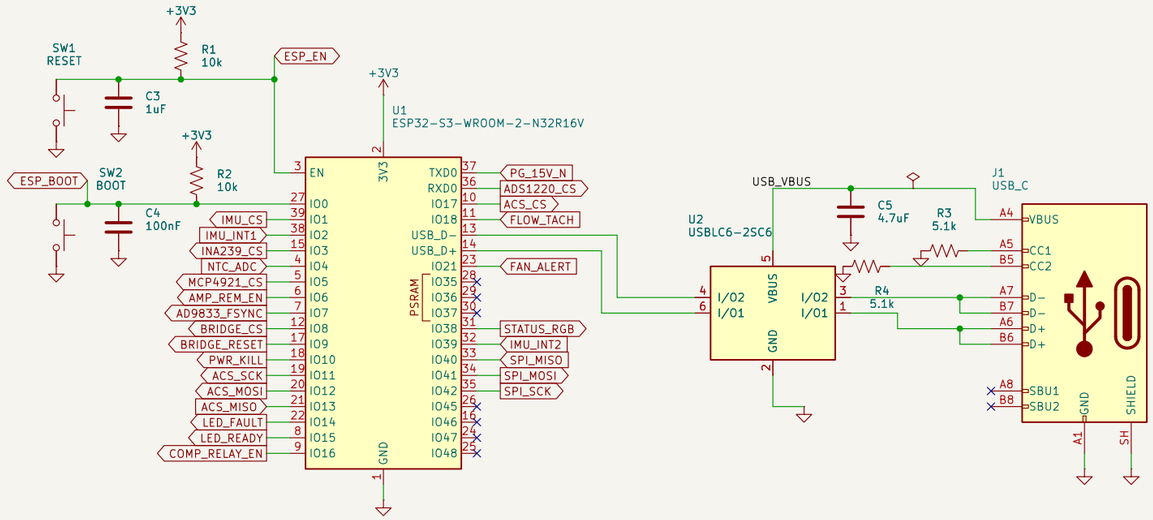
ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

GPIO map follows include/config/pin_config.h

## ESP32-S3 DECOUPLING CAPACITORS




Power Input
File: sheets/power_input.kicad_sch

Signal Generator
File: sheets/signal_gen.kicad_sch

Water Cooling
File: sheets/cooling.kicad_sch

Power Button
File: sheets/power_button.kicad_sch

Amp Control
File: sheets/amp_control.kicad_sch

Cold Head Temp
File: sheets/cold_head.kicad_sch

Rails 5V 3V3
File: sheets/rails_logic.kicad_sch

Amp Current
File: sheets/amp_current.kicad_sch

IMU
File: sheets/imu.kicad_sch

Rails +-15V
File: sheets/rails_15v.kicad_sch

Power Good
File: sheets/power_good.kicad_sch

System Current
File: sheets/system_current.kicad_sch

Indicators
File: sheets/indicators.kicad_sch

Compressor Relay
File: sheets/compressor_relay.kicad_sch

Water Cooling I2C Bridge
File: sheets/i2c_bridge.kicad_sch

Debug
File: sheets/debug.kicad_sch

Supply Switches

**Cryocooler Controller**

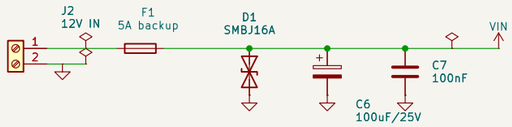
Sheet: /
File: cryocooler.kicad_sch

***Title: Cryocooler Controller***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/18 |

12 V input from a Dell XYK93 (DPS-2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

F1 is a backup only: add an inline high-breaking-capacity fuse (7.5-10 A) at the PSU end. Reverse polarity: TVS forward-clamps and blows the fuse.




**Cryocooler Controller**

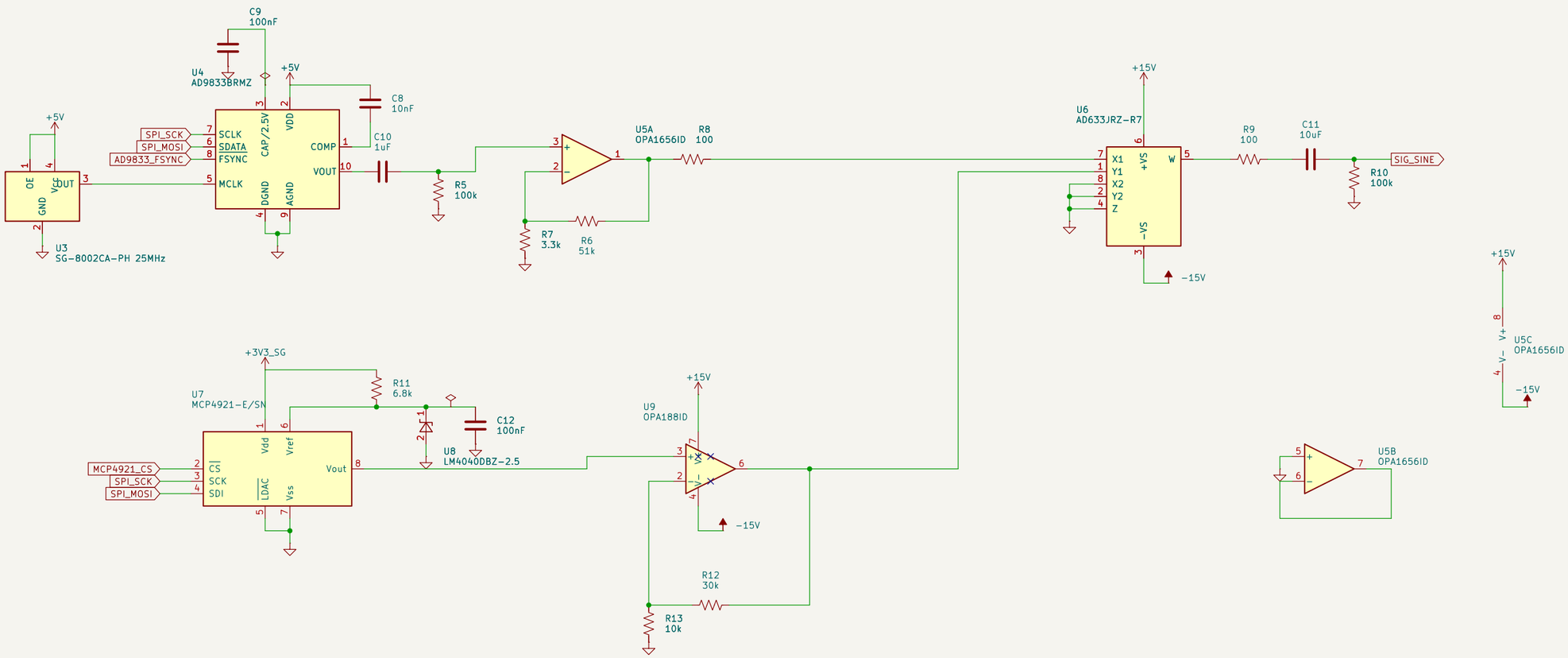
Sheet: /Power Input/
File: power_input.kicad_sch

***Title: Power Input***

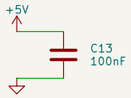
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/18 |

AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0–10 V amplitude control.

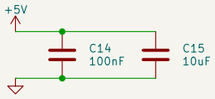
Vctl 0–10 V (2.5 V ref x4) x sine +/–4.9 V pk / 10 = 0..4.9 V pk (9.9 Vpp, 3.5 Vrms) to amplifier input; ZE500.1 RCA input range 0.2–6 V.



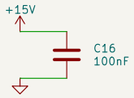

## SG–8002CA DECOUPLING CAPACITORS



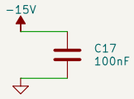

## AD9833 DECOUPLING CAPACITORS



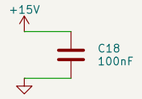

## OPA1656 +15V DECOUPLING CAPACITORS



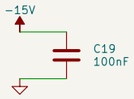

## OPA1656 –15V DECOUPLING CAPACITORS



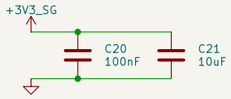

## AD633 +15V DECOUPLING CAPACITORS



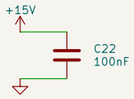

## AD633 –15V DECOUPLING CAPACITORS



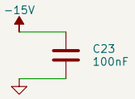

## MCP4921 DECOUPLING CAPACITORS




## OPA188 +15V DECOUPLING CAPACITORS




## OPA188 –15V DECOUPLING CAPACITORS



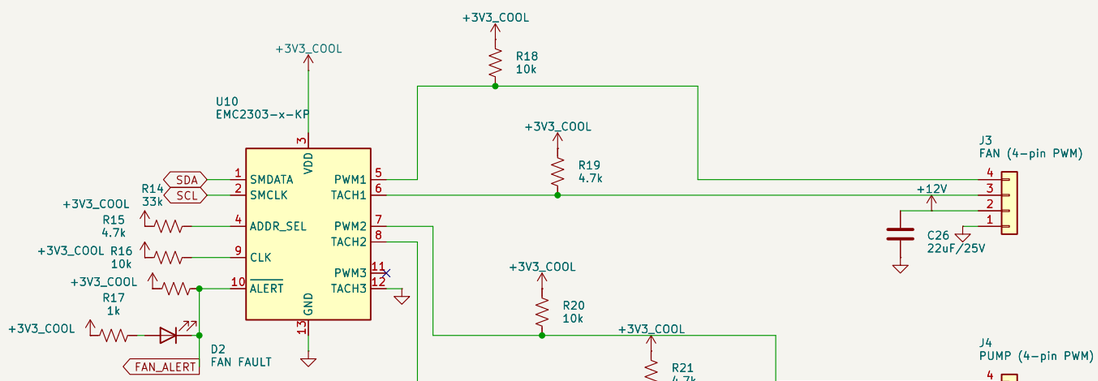

**Cryocooler Controller**

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

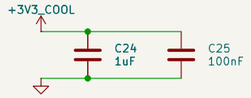
***Title: Signal Generator***

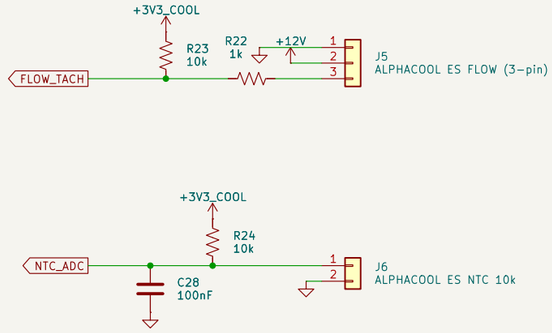
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/18 |

EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.




EMC2303 DECOUPLING CAPACITORS




**Cryocooler Controller**

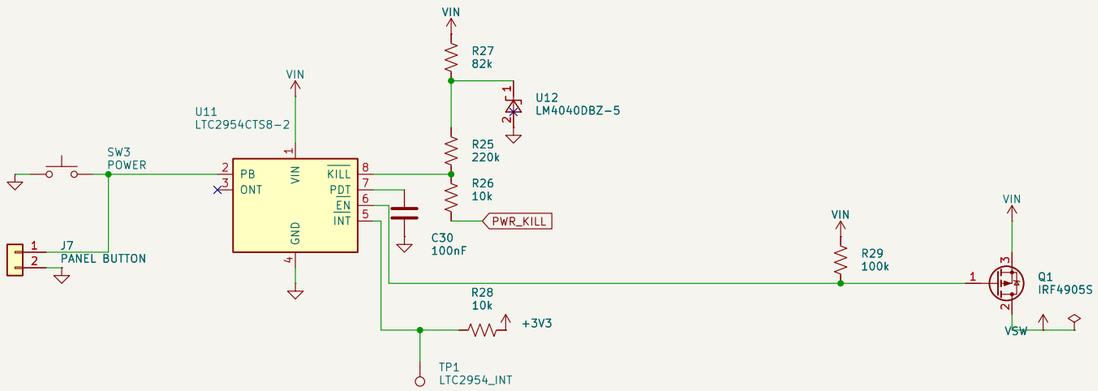
Sheet: /Water Cooling/
File: cooling.kicad_sch

***Title: Water Cooling***

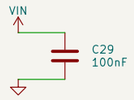
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/18 |

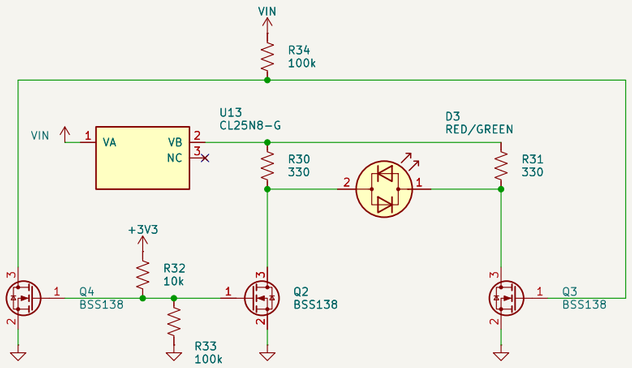
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby




LTC2954 DECOUPLING CAPACITORS

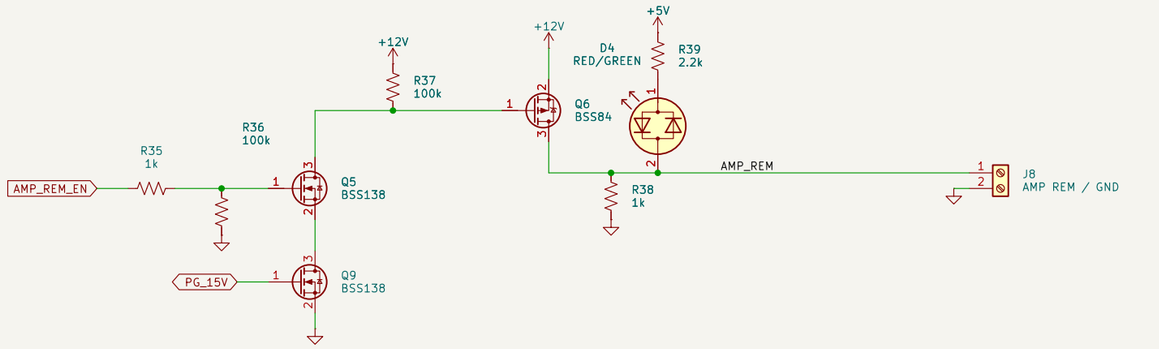



| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Power Button/ | | |
| File: power_button.kicad_sch | | |
| ***Title: Power Button*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 5/18 |

GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch), but only while PG_15V is high (both +-15 V rails good: Q9 in series with the level shifter). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.







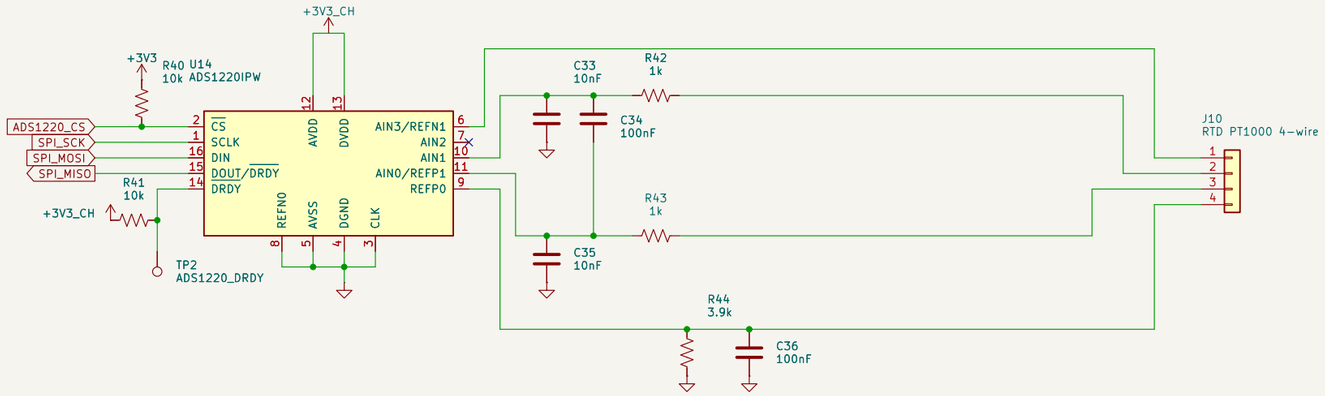
**Cryocooler Controller**

Sheet: /Amp Control/
File: amp_control.kicad_sch

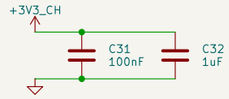
***Title: Amp Control***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/18 |

PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Rref (3.9k) on REFP0/REFN0. SPI mode 1 on the main bus (CS GPIO44), internal oscillator (CLK tied low). DRDY is not wired to the MCU.




## ADS1220 DECOUPLING CAPACITORS




**Cryocooler Controller**

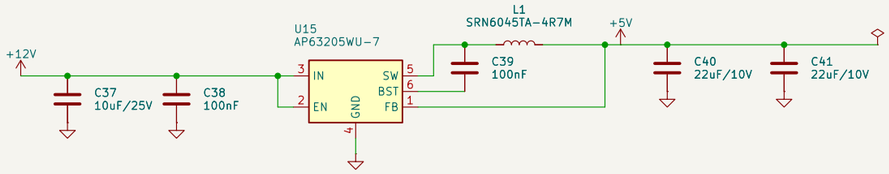
Sheet: /Cold Head Temp/
File: cold_head.kicad_sch

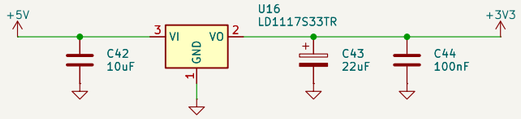
***Title: Cold Head Temp***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/18 |

12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.




**Cryocooler Controller**

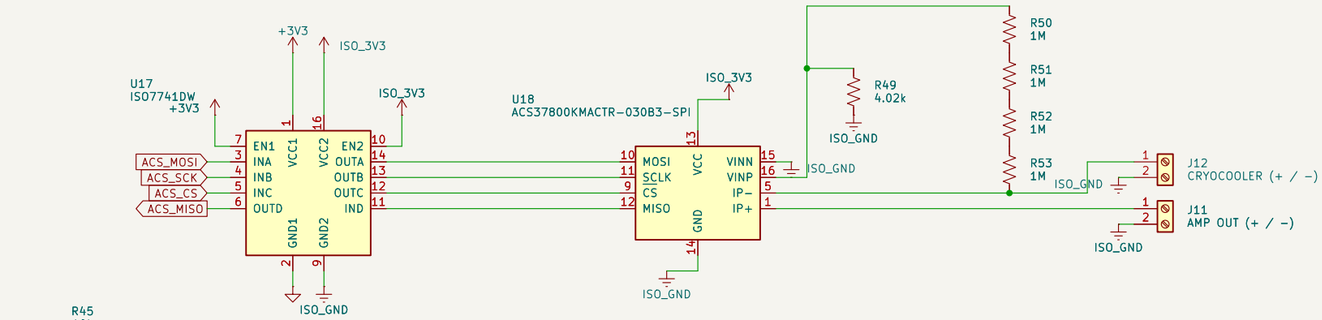
Sheet: /Rails 5V 3V3/
File: rails_logic.kicad_sch

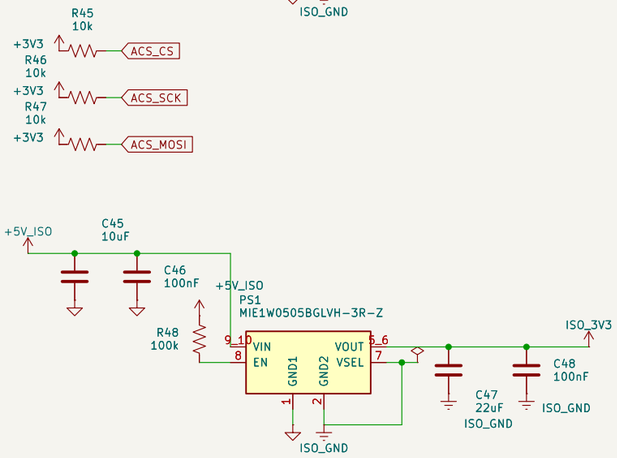
***Title: Rails 5V 3V3***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/18 |

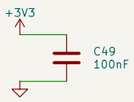
Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.

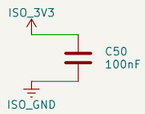


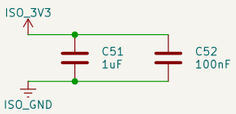

ISO7741 VCC1 DECOUPLING CAPACITORS




ISO7741 VCC2 DECOUPLING CAPACITORS




ACS37800 DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /Amp Current/

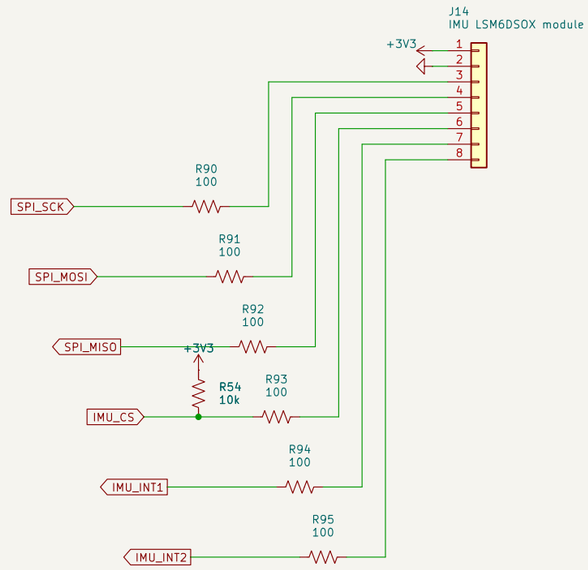
File: amp_current.kicad_sch

***Title: Amp Current***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 9/18 |

LSM6DSOX is the off-board Adafruit module (product 4438) on the cooler, cabled to J14 over the main SPI bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).

Module Vin = +3V3 keeps its logic at 3.3 V (its 3Vo pad stays open). 100 R series resistors damp the cable stub; the 10 k keeps CS high with the cable unplugged.




**Cryocooler Controller**

Sheet: /IMU/
File: imu.kicad_sch

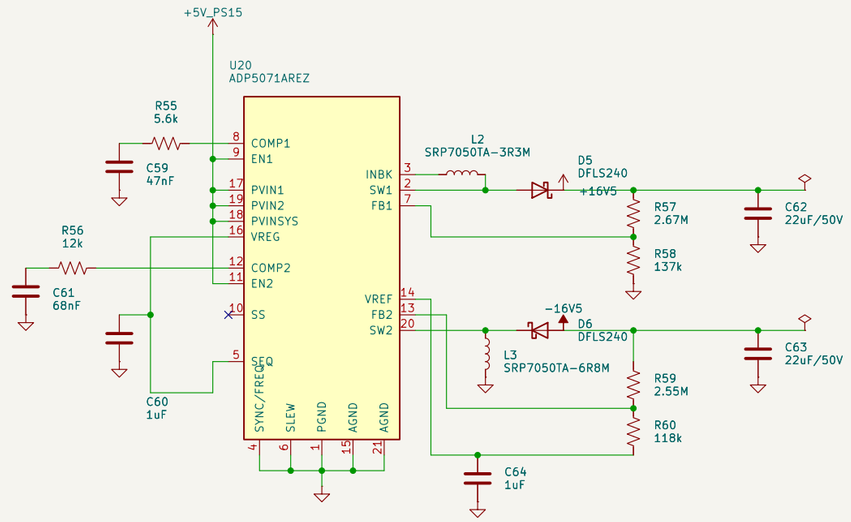
***Title: IMU***

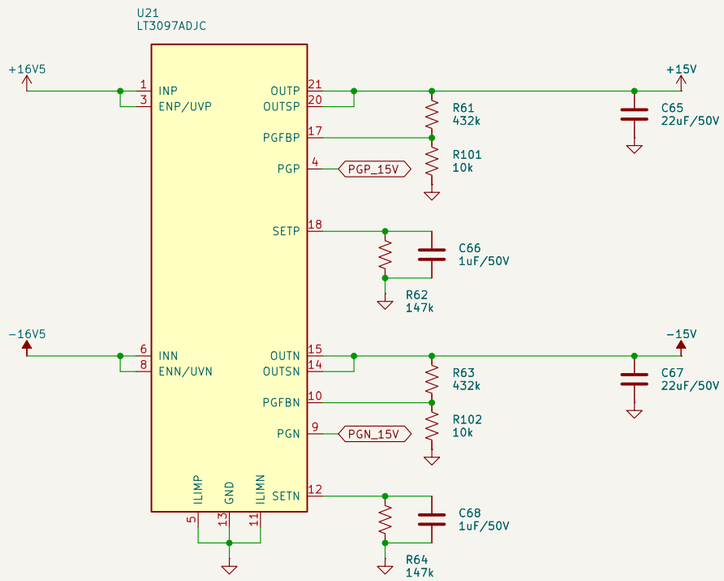
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/18 |

ADP5071AREZ (5 V in): +16.4 V boost and −16.5 V inverting rails; one LT3097 post-regulates them to +/−14.7 V.

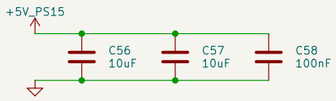
ADP5071 values from datasheet Fig. 47 / Tables 9–11. LT3097: VOUT = 100 uA x RSET (147k = 14.7 V, +15 V is the top of its range); PGFB dividers 432k/10k trip power good near 13.3 V (also enables fast start-up).

PGP and PGN (open collector, low = bad) go to nodes PGP_15V / PGN_15V built on the power_good sheet, which also makes PG_15V and lights one LED per pin.




ADP5071 DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /Rails +−15V/
File: rails_15v.kicad_sch

***Title: Rails +−15V***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 11/18 |

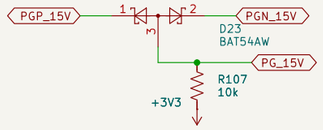
PGP and PGN pull their own node low when bad. Each node is the +15 V rail through 220k/62k (about 3.2 V when good, 0 V if its pin pulls low or the rail is missing) and lights its green LED.

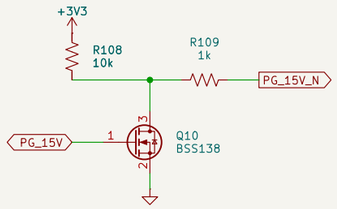
PG_15V = the two nodes ANDed by a BAT54AW with a 10k pull-up to +3V3 (3.3 V only when both are good); it gates the amplifier REM switch and reads LOW on GPIO43 (PG_15V_N) when both rails are good.

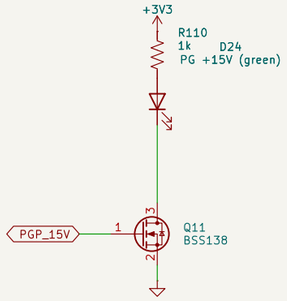


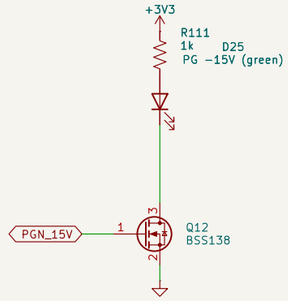






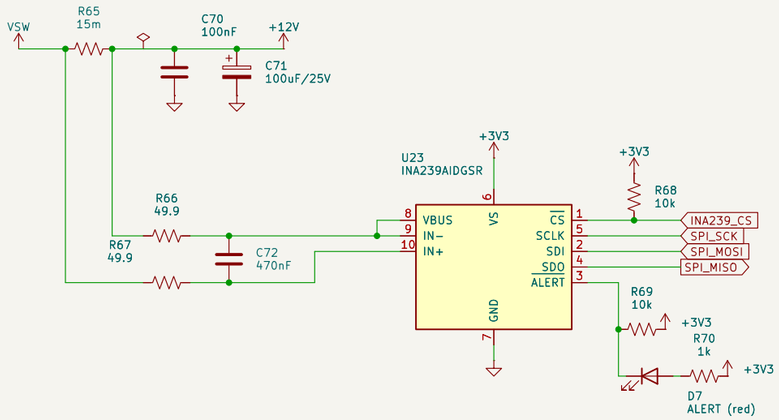

**Cryocooler Controller**

Sheet: /Power Good/
File: power_good.kicad_sch

***Title: Power Good***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 12/18 |

INA239 monitors the switched 12 V rail: 15 mOhm shunt, SPI mode 1 on the main bus (CS GPIO3), +/−163.84 mV range (~10.9 A). ALERT only drives its LED.




## 3V3 POWER LED




## INA239 DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /System Current/
File: system_current.kicad_sch

***Title: System Current***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 13/18 |

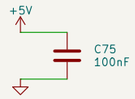
READY (GPIO15) and FAULT (GPIO14) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).



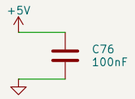


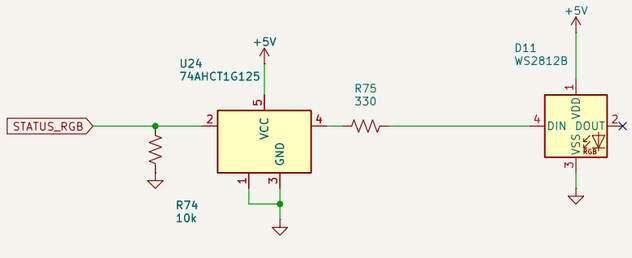


74AHCT1G125 DECOUPLING CAPACITORS



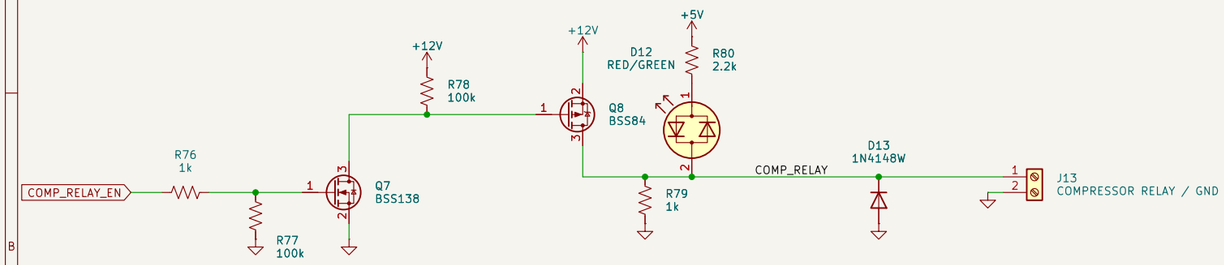

WS2812B DECOUPLING CAPACITORS




| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Indicators/<br>File: indicators.kicad_sch | | |
| ***Title: Indicators*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 14/18 |

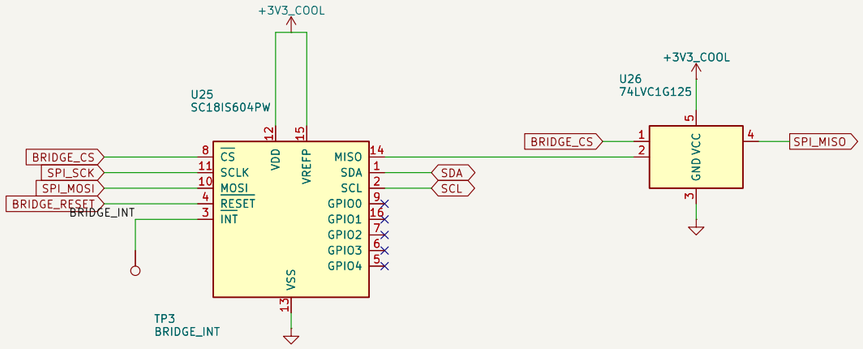
COMP_RELAY_EN (GPIO16) high -> +12 V on the compressor relay terminal (BSS84 high-side switch, 130 mA max: drive an external relay or SSR coil). Flyback diode across the output. LED: green = on, red = off.



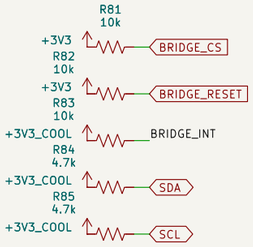

| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Compressor Relay/<br>File: compressor_relay.kicad_sch | | |
| ***Title: Compressor Relay*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 15/18 |

SC18IS604: SPI mode 3 slave on the main bus (CS GPIO8, RESET GPIO9), I2C controller at ~100 kHz for the EMC2303 (0x4D). INT# is polled through the I2CStat register and only has a pull-up and a test point.

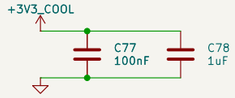
The datasheet does not say MISO is high-impedance with CS high, so a 74LVC1G125 (OE# = CS) gates it onto SPI_MISO.



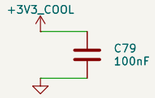

## PULL-UPS




## SC18IS604 DECOUPLING CAPACITORS




## 74LVC1G125 DECOUPLING CAPACITORS




**Cryocooler Controller**

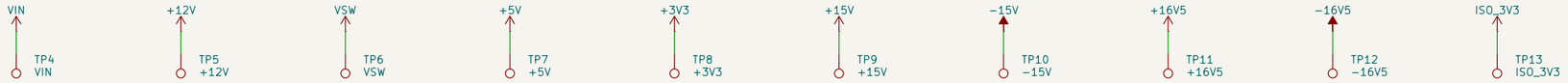
Sheet: /Water Cooling I2C Bridge/
File: i2c_bridge.kicad_sch

***Title: Water Cooling I2C Bridge***

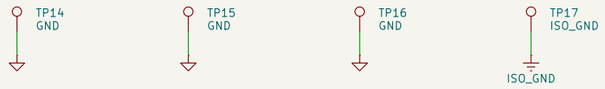
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 16/18 |

Debug aids: test points (pad per net, loops for ground) and rail LEDs. Each test point sits on a global label or power symbol; the PCB generator places it beside a pad on the same net.

## POWER RAILS




## GROUND (scope loops)




## SPI BUSES AND CHIP SELECTS



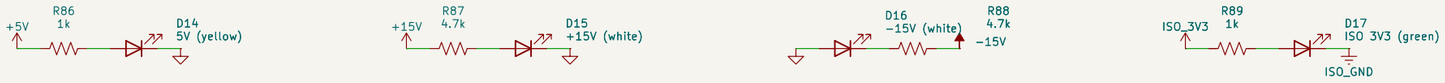

## I2C, INTERRUPTS AND CONTROL




## ANALOG AND SENSORS




## RAIL LEDS




**Cryocooler Controller**

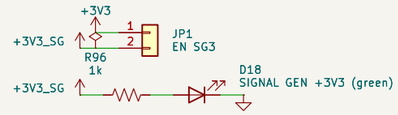
Sheet: /Debug/
File: debug.kicad_sch

***Title: Debug***

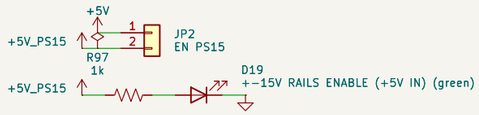
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 17/18 |

Each 1.27 mm jumper sits in series with one load supply: pull it to power that part off. The LED shows the switched rail is present. A part left unpowered can still be fed through its signal pins, so disconnect the cable too when you need it fully dead.

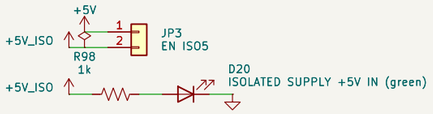
SIGNAL GEN +3V3



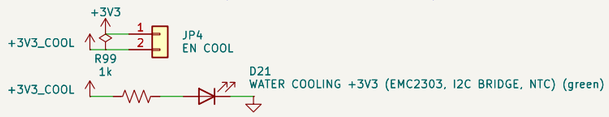

+−15V RAILS ENABLE (+5V IN)



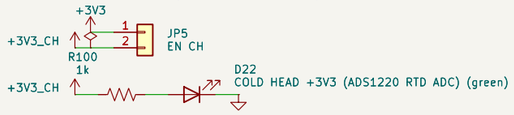

ISOLATED SUPPLY +5V IN




WATER COOLING +3V3 (EMC2303, I2C BRIDGE, NTC)




COLD HEAD +3V3 (ADS1220 RTD ADC)




**Cryocooler Controller**

Sheet: /Supply Switches/
File: supply_switches.kicad_sch

***Title: Supply Switches***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 18/18 |